Ryan Saelens
Comp. Phys.
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① Integral Problem → Uploaded to GitHub

② Single Precision Binary Rep. of $-1/3$

Sign Bit = 1 → negative

$0.3 \times 2 = 0.6$
$0.6 \times 2 = 1.2$
$0.2 \times 2 = 0.4$
$0.4 \times 2 = 0.8$
$0.8 \times 2 = 1.6$
$0.6 \times 2 = 1.2$
$0.2 \times 2 = 0.4$
$0.4 \times 2 = 0.8$
$0.8 \times 2 = 1.6$
⋮ Repeating ⋮

In Binary: 1111101

Exponent $0.01\,0011001 \times 2$

$1.\underbrace{0011001}_{\text{Mantissa}} \times 2^{-2}$; $-2+127 = 125$

Sign | Exponent | Mantissa
1 | 01111101 | 00110011000000...

③ Smallest Positive Integer : not exact : single prec.

- using $2^{n+1}+1$ → values allowed regarding mantissa
extra precision → mantissa consists of 23 bits

$\Rightarrow 2^{24}+1 = \underline{16777217}$ → So, integers upto 2^{23} are allowed
→ So if $n = 23$, $+1 \to 24$, will give first non-exact value

④ Two's Complements : 9, 455, 1021, -12, -1022, -1023

9 = 01001

2's Comp. 10110

+ 1

10111

-12 : find 12 first :

12 : 01100

2's Comp : 10011

+ 1

10100

1021 : find -1021 first

-(1024)	512	256	128	64	32	16	8	4	2	1
1	0	0	0	0	0	0	0	0	1	1

-1021 = 10000000011

→ 10000000010 (before adding 1)

→ therefore 1021 = 01111111101

-1022 : find 1022 first :

1022 : 01111111110 (using same method)

-1022 : 10000000001

+ 1

10000000010

-1023 : find 1023 first :

1023 : 01111111111

-1023 : 10000000000

+ 1

10000000001

(5) Binary, Octal, Hexidecimal Reps. of 13, 133, 1333, 13333

	B	O	HD
13	1101	15	D
133	10000101	205	85
1333	10100110101	2465	535
13333	11010000010101	32025	3415

& Work

133 ÷ 2	66	1
66 ÷ 2	33	0
33 ÷ 2	16	1
16 ÷ 2	8	0
8 ÷ 2	4	0
4 ÷ 2	2	0
2 ÷ 2	1	0
1 ÷ 2	0	1

133 ÷ 8	16	.5
	8 - 0.625	
16 ÷ 8	2	0
2 ÷ 8	0	2

133 ÷ 16	8	.5
8 ÷ 16	0	.8

1333 ÷ 2	666	1
666 ÷ 2	333	0
333 ÷ 2	166	1
166 ÷ 2	83	0
83 ÷ 2	41	1
41 ÷ 2	20	1
20 ÷ 2	10	0
10 ÷ 2	5	0
5 ÷ 2	2	1
2 ÷ 2	1	0
1 ÷ 2	0	1

1333 ÷ 8	166	5
166 ÷ 8	20	6
20 ÷ 8	2	4
2 ÷ 8	0	2

1333 ÷ 16	83	5
83 ÷ 16	5	3
5 ÷ 16	0	5

13333 ÷ 2	6666	1
6666 ÷ 2	3333	0
3333 ÷ 2	1666	1
1666 ÷ 2	833	0
833 ÷ 2	416	1
416 ÷ 2	208	0
208 ÷ 2	104	0
104 ÷ 2	52	0
52 ÷ 2	26	0
26 ÷ 2	13	0
13 ÷ 2	6	1
6 ÷ 2	3	0
3 ÷ 2	1	1
1 ÷ 2	0	1

13333 ÷ 8	1666	5
1666 ÷ 8	208	2
208 ÷ 8	26	0
26 ÷ 8	3	2
3 ÷ 8	0	3

13333 ÷ 16	833	5
833 ÷ 16	52	1
52 ÷ 16	3	4
3 ÷ 16	0	3